Swiggy — Descriptive Analytics Report

(Simulated Data)

Order Volume & Customer Behavior at Scale

Tracking Most-Ordered Restaurants, Ordering Trends, Ratings, City-Wise Demand, Customer Behavior & Restaurant Partner Health

Swiggy — Simulated Food Delivery Platform Dataset

Report window: 30 days | 60 Restaurant Partners | 10 cuisines | 8 cities

Table of Contents

1. Note on Data & Objective
2. How This Would Work at Real Scale
3. Daily Ordering Trend
3.1 Daily Platform Revenue Trend
4. Most-Ordered Restaurants (Most Recent Day)
4.1 Most-Ordered Restaurants by New Customer Reach
5. Rating Distribution
5.1 Restaurants with Declining Rating Trend
5.2 High-Risk Combination: High Order Volume, Sub-3.5 Rating
6. City-Wise Demand
6.1 Cuisine-Wise Performance
6.2 Cuisine Share of Total Orders
6.3 Day-of-Week Ordering Pattern
6.4 Device / Platform Mix
6.5 Order Status & Cancellations by Cuisine
6.6 Swiggy One Membership Adoption by City
6.7 Revenue by Membership Tier
7. Restaurant Partner Status
7.1 Partners Nearing Contract Renewal — Renewal Risk
8. Summary of Key Metrics
9. Next Steps to Move From Descriptive → Predictive
Appendix A. Simulation Methodology
Appendix B. Cuisine Reference Table

1. Note on Data & Objective

Swiggy's live order and customer data isn't publicly accessible, so this report uses a synthetic dataset built to mirror real food-delivery platform patterns (60 restaurant partners across 10 cuisines, 8 cities, 30 days, ~20.3 million simulated orders). It is designed to show exactly what a descriptive analytics pipeline processing millions of daily order events would produce — the same structure applies once real data is substituted in.

Objective

Build a descriptive analytics layer over daily ordering and customer activity to answer six recurring business questions:

- What is being ordered most right now, and by how many customers?
- How is total order volume and revenue trending day over day?
- How are restaurants rated, and where is service quality slipping?
- Where is order demand concentrated geographically, and by cuisine?
- How does customer behavior — membership tier, device, cancellations — vary by city and cuisine?
- Which restaurant partners are at risk of losing their platform contract before demand fades?

2. How This Would Work at Real Scale

This report was actually produced with a lightweight, practical toolchain — the same one a solo analyst or a small team would reach for before any big-data infrastructure is justified:

Layer	Purpose	Typical Tools
Data source	Raw order/customer records, exported on a schedule	Excel (.xlsx) / CSV export from the source system
Storage & querying	Load the export into a queryable table, join/filter/aggregate	Python
Analysis	Clean, reshape, and compute the metrics behind every chart in this report	Python (pandas, numpy) in a Jupyter Notebook
Visualization	Generate each figure from the aggregated tables	Python (matplotlib)
Reporting	Assemble the narrative, tables, and charts into a shareable document	Notebook export / scripted PDF report

This stack scales further than it looks: SQL handles the aggregation once row counts get too large for Excel alone, and the same pandas/matplotlib notebook logic can be pointed at a warehouse table instead of a CSV without changing the analysis itself. The jump to a Kafka/Spark/Redshift-style pipeline only becomes necessary once ingestion needs to happen continuously, in real time, at millions of order events per day — for a daily or weekly reporting cadence like this one, Excel/SQL/Python/Jupyter is the right-sized tool, not an early version of a bigger system.

The report that follows reflects the output of that workflow — daily aggregated tables computed in pandas and charted with matplotlib — not raw event-level data, since no report renders millions of individual rows usefully.

3. Daily Ordering Trend

Over the simulated 30-day window: ~20.3 million orders placed, generating ~Rs. 577 Crore in gross order value, with a weekly seasonal cycle (mid-week dips, weekend peaks) layered on top of gradual cuisine-level growth or decline trends per restaurant.

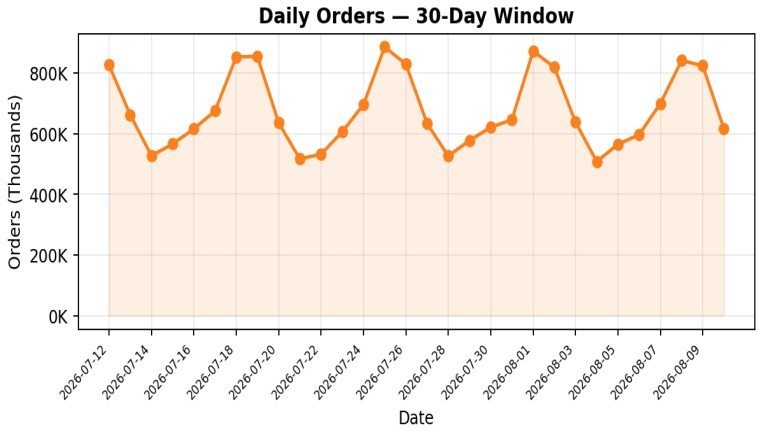


Figure 1. Daily orders across the 30-day window. Weekend peaks and mid-week dips form a repeating cycle.

Key Insight: Ordering is highly cyclical, not growing — order volume oscillates on a repeating weekly pattern, peaking on weekends, with no clear upward or downward trend over the 30 days.

KPI Definitions Used

- **Daily orders** = count of all placed orders (delivered + cancelled) across the platform that day
- **Daily platform revenue** = commission/take-rate revenue earned on gross order value for that day, net of the aggregation layer (cancellations net out separately)

3.1 Daily Platform Revenue Trend

Revenue broadly tracks the order-volume curve but is not identical to it — days with heavier ordering from Swiggy One members or high-AOV cuisines (Pizza, Biryani) push revenue up even without a matching spike in order count.

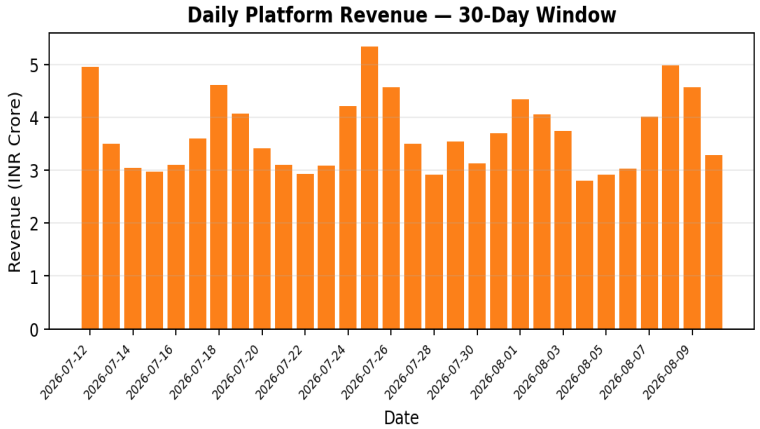


Figure 2. Daily platform revenue in INR Crore. Compare the shape against Figure 1 — divergence signals a shift in AOV or cuisine mix.

Key Insight: Revenue doesn't track orders 1:1 — some lower-volume days out-earn higher-volume ones, pointing to shifts in average order value (premium cuisines, membership mix) rather than raw order count.

4. Most-Ordered Restaurants (Most Recent Day)

Top performer led with 94,675 orders in a single day. This is a common "hero restaurant" pattern on food-delivery platforms: a small number of restaurants drive a disproportionate share of ordering — a Pareto effect worth flagging for promotional placement and kitchen/logistics capacity planning.

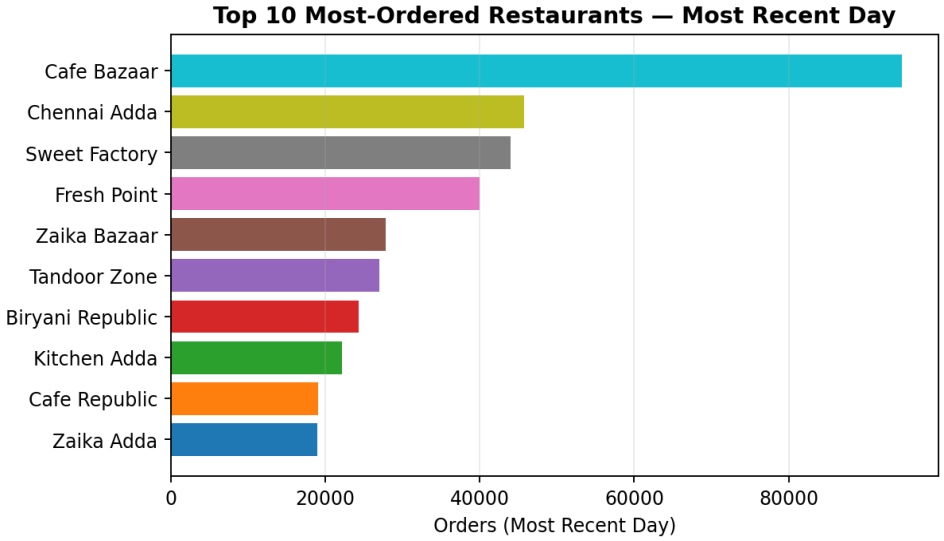


Figure 3. Top 10 restaurants by orders on the most recent day.

> **Key Insight:** One hero restaurant dominates — "Cafe Bazaar" (94,675 orders) leads by a wide margin over #2, a classic Pareto pattern worth flagging for delivery-fleet capacity planning.

Ranking logic: orders, not reach — a restaurant with fewer customers but a very high repeat-order rate can outrank a restaurant with more, one-time customers, so both cuts (orders and reach) should exist as separate leaderboard views in a real dashboard.

4.1 Most-Ordered Restaurants by New Customer Reach

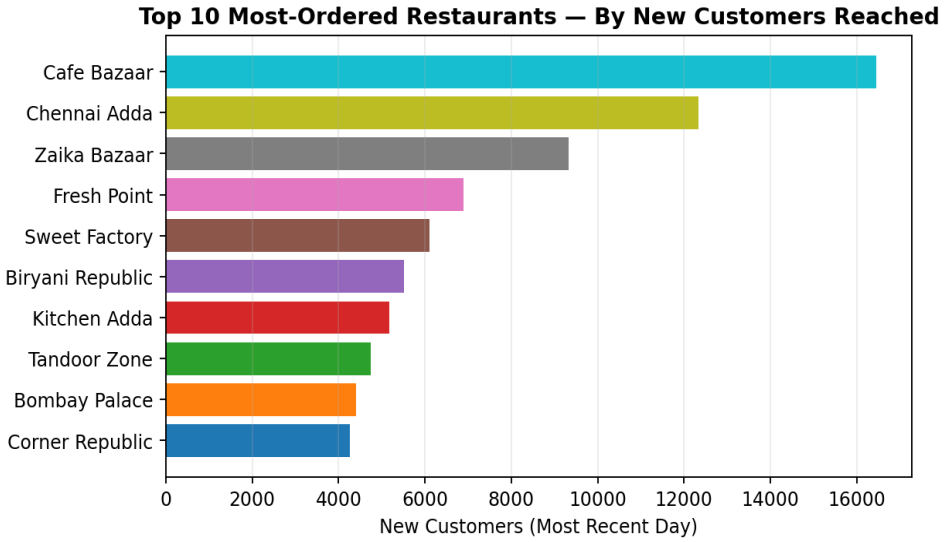


Figure 4. Top 10 restaurants by new customers reached on the most recent day — a different leaderboard from Figure 3.

Key Insight: Reach and volume rankings diverge — some restaurants rank high on new customers but not on the orders list, meaning some restaurants pull in breadth (many new customers) while others pull depth (repeat orders from existing customers).

Restaurant	Cuisine	New Customers
Cafe Bazaar	Chinese	16,453
Chennai Adda	Fast Food	12,326
Zaika Bazaar	Pizza	9,335
Fresh Point	Pizza	6,886
Sweet Factory	Beverages	6,109
Biryani Republic	Pizza	5,516

5. Rating Distribution

Average rating across all restaurant-days: 3.99 / 5, clustered tightly (most mass between 3.6 and 4.5). In a real system, this view is most useful segmented, not aggregated.

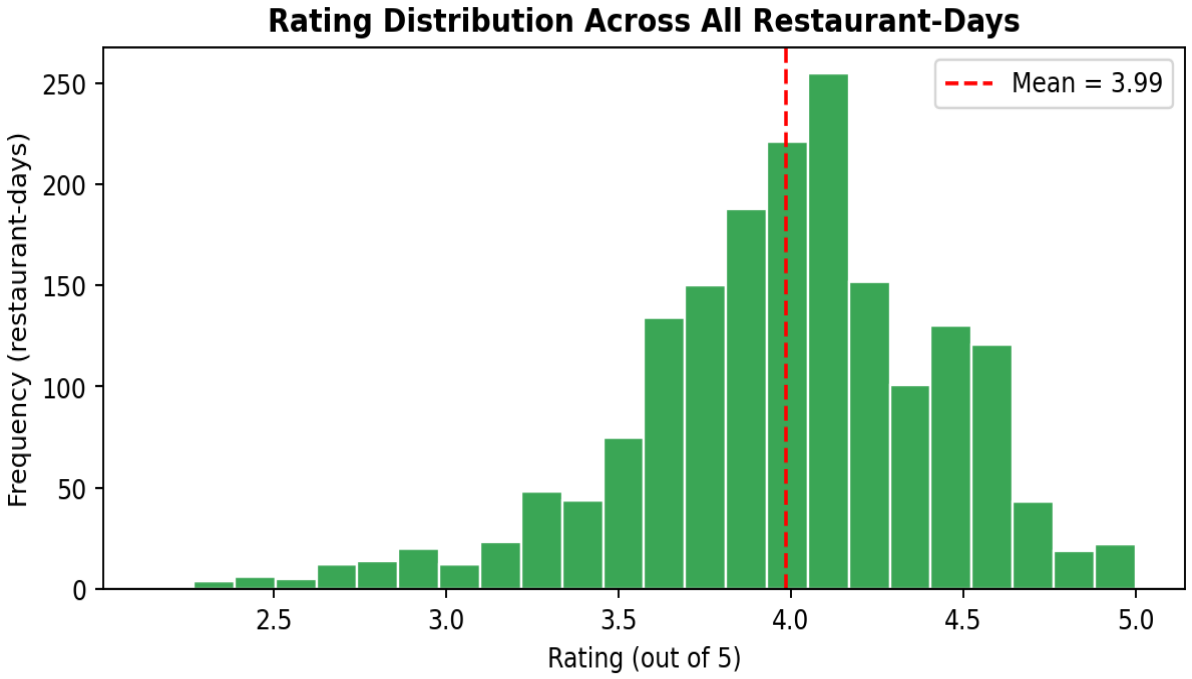


Figure 5. Distribution of ratings across all restaurant-days in the window.

Key Insight: Ratings cluster tightly and skew positive — mean 3.99/5, most mass between 3.6–4.5, so a single average score hides meaningful outliers.

• **Restaurants trending downward** in rolling 7-day average rating — the earliest signal of a food-quality, packaging, or delivery-time issue

• **Restaurants with high order volume but sub-3.5 ratings** — the highest-risk combination, since volume amplifies the reach of poor reviews and churn

5.1 Restaurants with Declining Rating Trend

6 restaurants show a sustained downward drift in their 7-day rolling average rating over the window — the earliest, cheapest signal that something in food quality, packaging, or delivery reliability has changed before ratings collapse at scale.

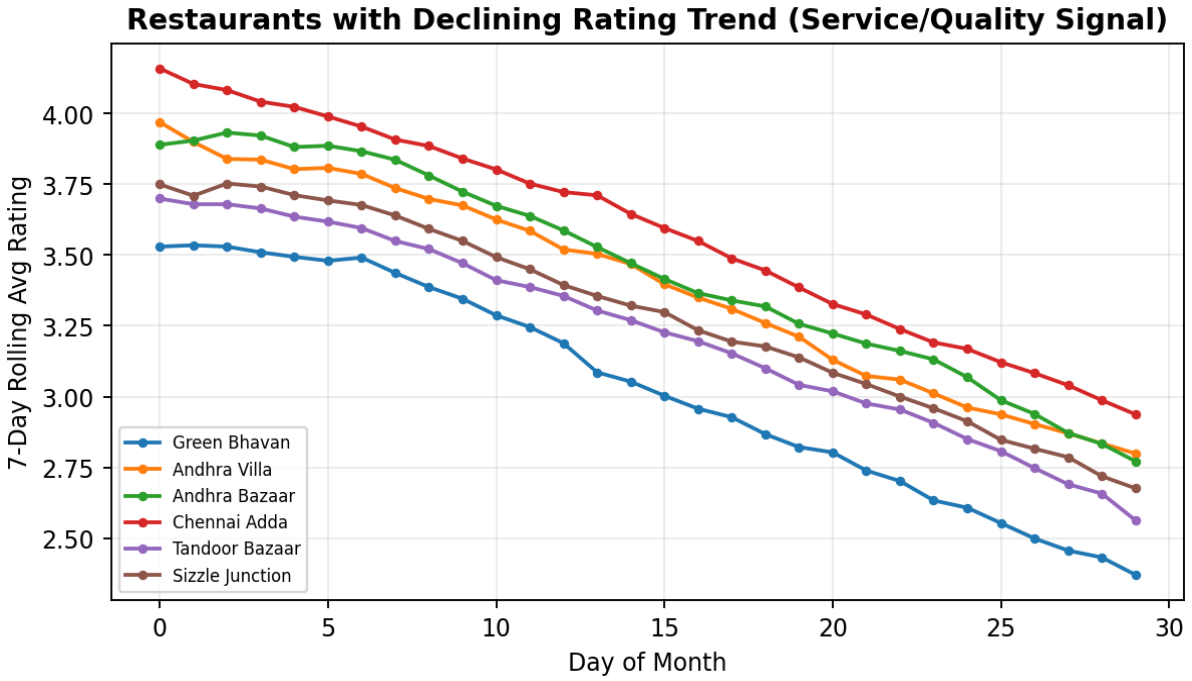


Figure 6. Rolling 7-day average rating for restaurants flagged with a declining trend.

Key Insight: 6 restaurants are bleeding quality signal — the steepest drops trend toward ~2.5–3.0, an early warning ahead of any ratings collapse.

Restaurant ID	Restaurant
SWGY-2033	Green Bhavan
SWGY-2036	Andhra Villa
SWGY-2042	Andhra Bazaar
SWGY-2044	Chennai Adda
SWGY-2050	Tandoor Bazaar
SWGY-2053	Sizzle Junction

5.2 High-Risk Combination: High Order Volume, Sub-3.5 Rating

These restaurants combine above-median monthly order volume with a current rating below 3.5 — the highest-risk combination on the platform, since every additional order reaches more customers who may then leave the same critical feedback or churn to a competitor.

Restaurant	Monthly Orders	Rating
Tandoor Bazaar	193,199	2.35
Sizzle Junction	312,116	2.57
Andhra Villa	193,148	2.72
Chennai Adda	1,462,883	2.83
Tandoor Zone	878,030	3.34

Key Insight: "Tandoor Bazaar" is the single biggest risk on the platform — 193,199 monthly orders at a 2.35 rating means poor reviews are reaching the widest possible audience.

Recommended action: route these restaurants to a manual food-quality / kitchen-audit review queue before the next in-app promotional push increases their exposure further.

6. City-Wise Demand

Order distribution across cities was fairly balanced in this simulation. In practice, city-level demand at Swiggy's real scale typically shows heavier concentration in a handful of core metros (Bangalore, Delhi NCR, Mumbai, Hyderabad), with the rest long-tailed across smaller or emerging cities.

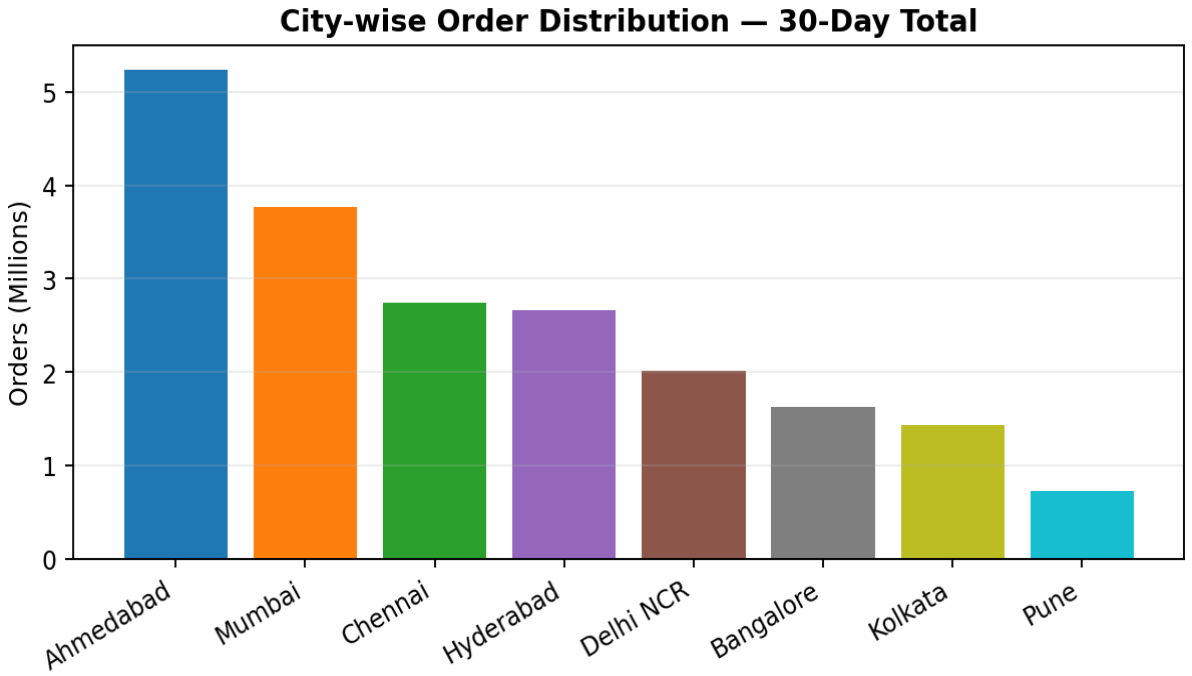


Figure 7. Orders by city over the 30-day window.

Key Insight: Ahmedabad leads order volume — demand is concentrated, not evenly spread, which should anchor delivery-fleet and dark-kitchen investment priority.

This view is what drives:

- City-level restaurant onboarding and dark-kitchen investment
- Localized offers, surge-pricing, and marketing spend
- Delivery-fleet capacity planning (demand spikes in cities with fewer active delivery partners)

6.1 Cuisine-Wise Performance

Layering cuisine onto the city view surfaces where revenue concentration actually sits, independent of raw order counts.

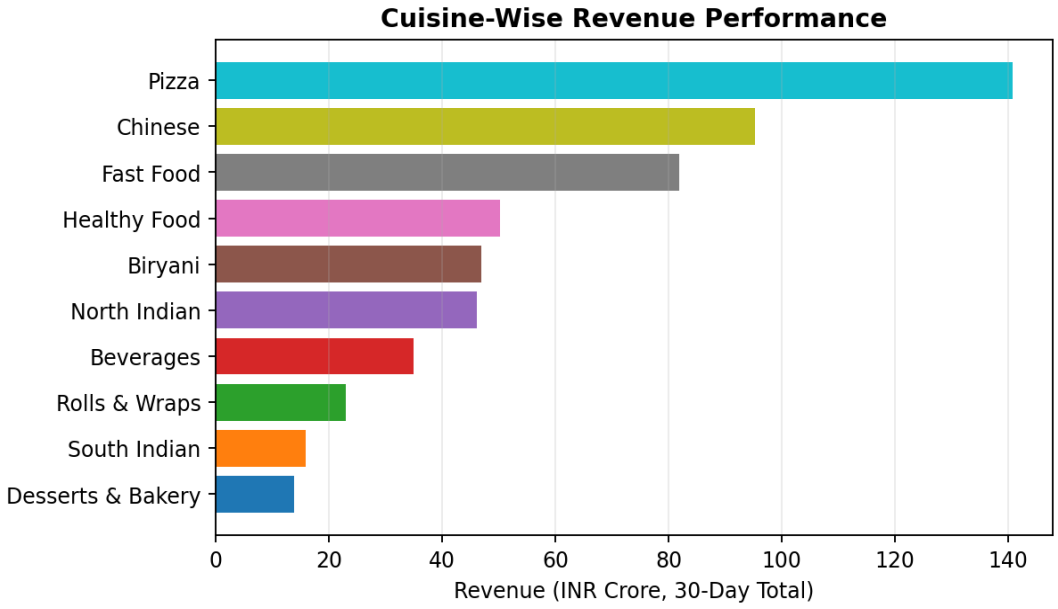


Figure 8. Revenue by cuisine, 30-day total, ranked highest to lowest.

Key Insight: Pizza leads revenue, notably outperforming its order-volume share — meaning cuisine mix and average order value, not just order count, drives platform revenue.

6.2 Cuisine Share of Total Orders

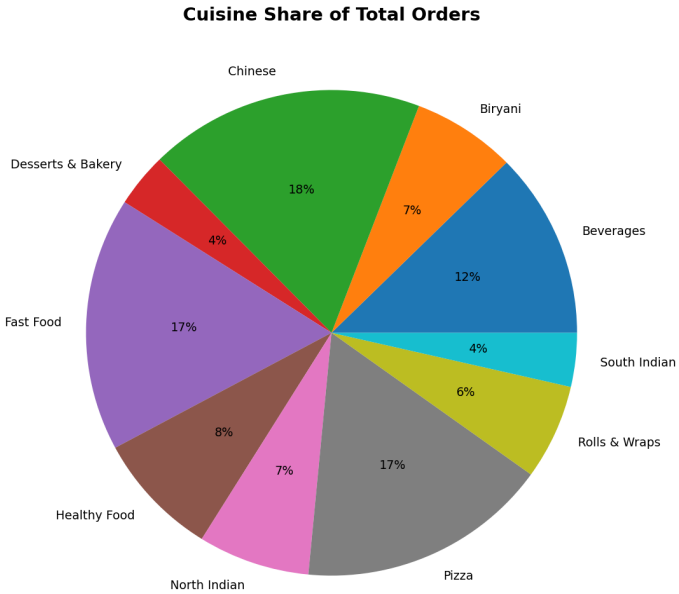


Figure 9. Share of total orders by cuisine.

Key Insight: Chinese, Fast Food, and Pizza form the volume backbone of the platform, while South Indian and Desserts & Bakery are more niche, suggesting a long-tail cuisine library propped up by a handful of core categories.

6.3 Day-of-Week Ordering Pattern

Aggregating the 30-day window by weekday isolates the seasonal cycle seen in Figure 1. Saturday is the strongest day for ordering, while Tuesday is consistently the softest — a pattern that should directly inform restaurant-partner staffing, homepage merchandising, and delivery-fleet scheduling.

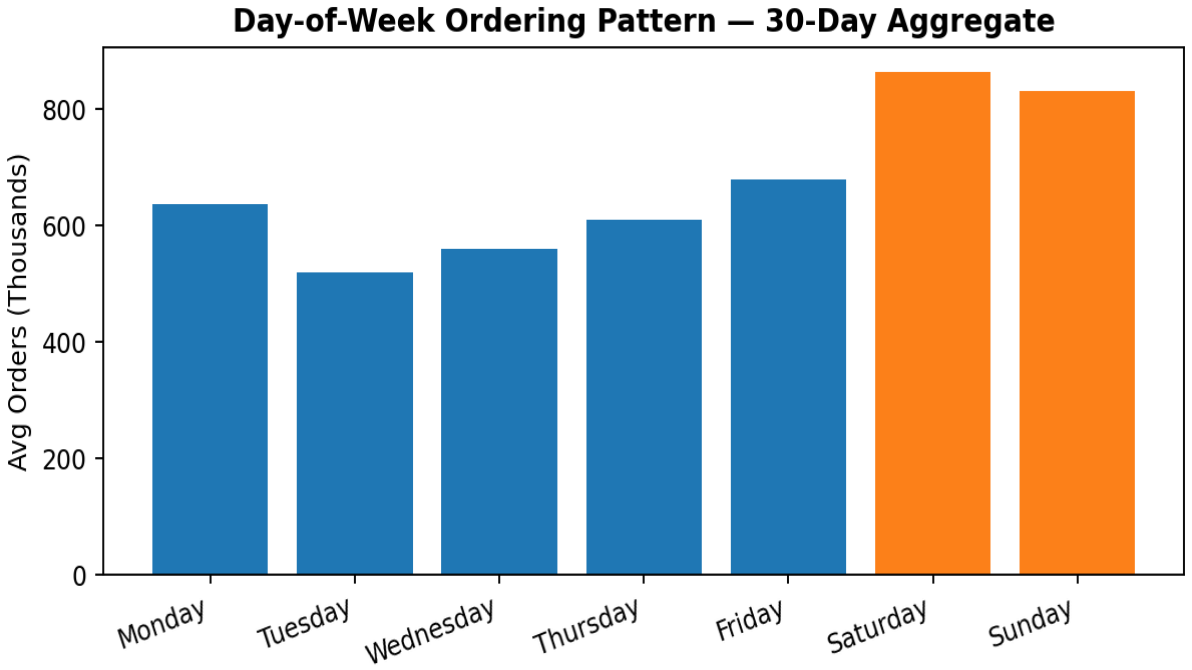


Figure 10. Average orders by day of week (weekend days highlighted).

Key Insight: Weekends convert to noticeably higher order volume — Saturday clearly outperforms the flat Mon–Fri baseline, with Tuesday the softest day — a strong signal for delivery-partner shift planning.

6.4 Device / Platform Mix

The Android app leads order share at 52%, consistent with Android's dominant smartphone share in Swiggy's core markets. Device mix matters operationally: web and voice-order sessions carry different conversion and payment-flow characteristics than native app sessions.

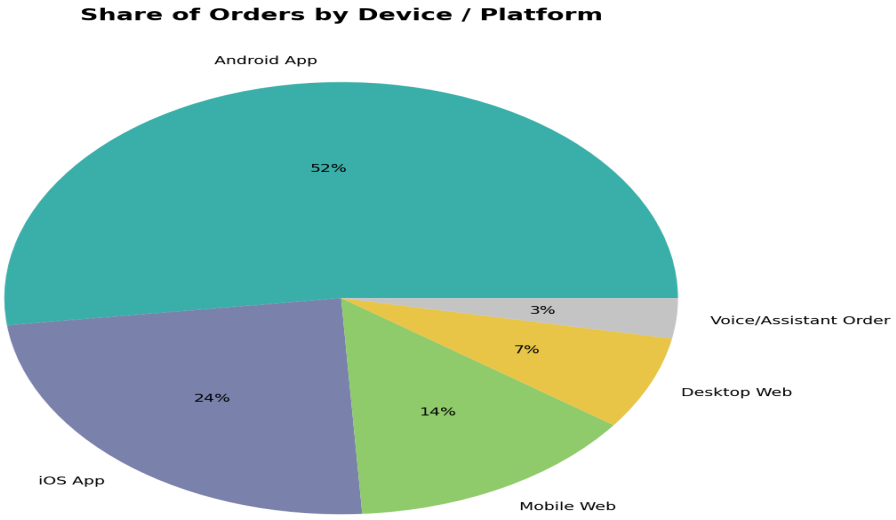


Figure 11. Share of orders by device / platform.

Key Insight: The Android app is the primary ordering surface at 52%, more than iOS and web combined trailing individually — confirms the mobile app, not desktop web, drives the platform's core engagement.

Device / Platform	Order Share	Orders
Android App	52%	10,538,545
iOS App	24%	4,863,944
Mobile Web	14%	2,837,300
Desktop Web	7%	1,418,650
Voice/Assistant Order	3%	607,993

6.5 Order Status & Cancellations by Cuisine

Cancellation rates vary meaningfully by cuisine — Biryani runs highest at ~9.5% cancelled, typical for longer-prep, high-demand items where kitchens hit capacity limits, while Healthy Food sits lowest at ~4.0%, as expected for simpler, purpose-sought orders. Cancellation and late-delivery rates should be tracked as a leading indicator alongside the rating metrics in Section 5.

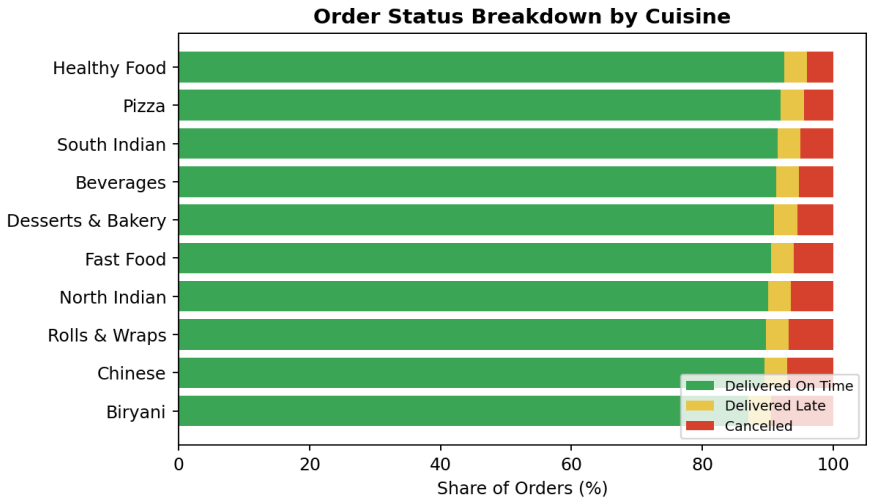


Figure 12. Order status composition by cuisine (Delivered On Time / Delivered Late / Cancelled).

Key Insight: Biryani has the worst cancellation rate (~9.5%) while Healthy Food holds up best (~4.0% cancelled) — cuisine-level cancellation should be tracked alongside ratings as a leading indicator.

6.6 Swiggy One Membership Adoption by City

Bangalore carries the deepest Swiggy One adoption at 41%, typical of mature, high-frequency-ordering metros, while Kolkata stays leanest at 15% — cities with greater price sensitivity or a newer membership push.

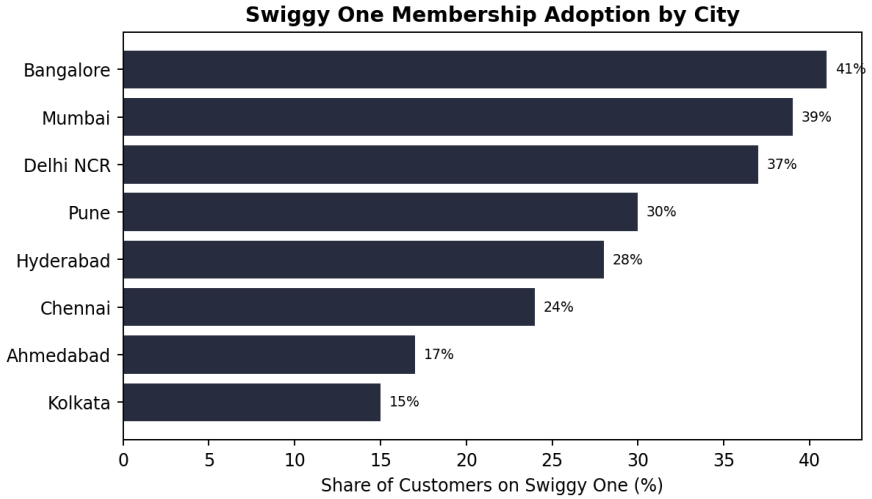


Figure 13. Swiggy One membership adoption by city, sorted highest to lowest.

Key Insight: A wide gap in Swiggy One adoption exists between markets — Bangalore (41%) vs. Kolkata (15%), reflecting frequency and price-sensitivity differences that should inform regional membership strategy.

6.7 Revenue by Membership Tier

Segmenting gross order value by membership tier shows Swiggy One members contributing the larger share of platform order value — consistent with free delivery and exclusive-deal access driving higher ordering frequency. This cut is what separates an "orders" story from a "revenue" story when planning membership-pricing changes.

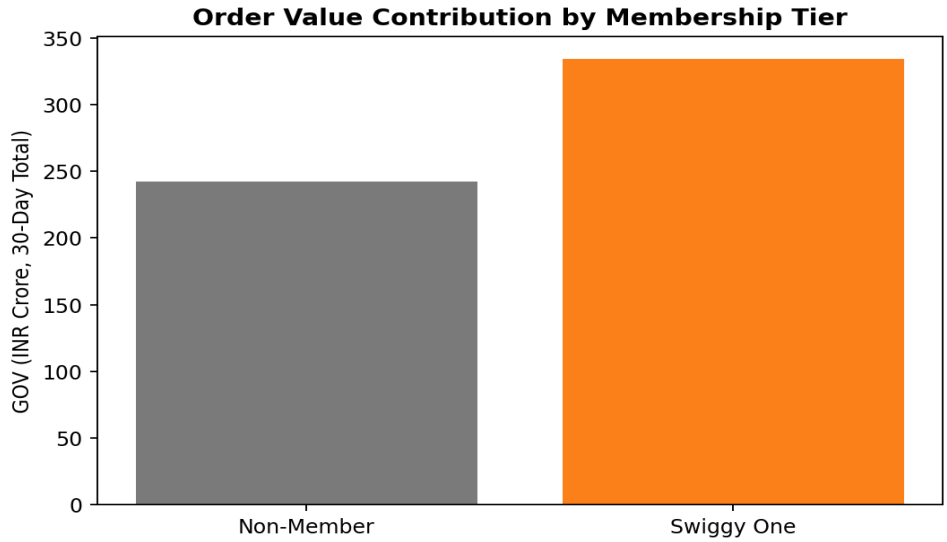


Figure 14. Gross order value contribution by membership tier across the 30-day window.

Key Insight: Swiggy One members drive 58% of gross order value from a smaller subscriber base — membership conversion has an outsized effect on order frequency and per-customer value.

Membership Tier	GOV Share	Gross Order Value (INR)
Non-Member	42%	Rs. 2,421,394,161
Swiggy One	58%	Rs. 3,343,830,031

7. Restaurant Partner Status

Restaurant partner contracts were bucketed into three tiers:

- **Active Partnership** — comfortable runway before contract renewal is needed
- **Renewal Due Soon** (< 30 days) — renewal-decision trigger zone
- **Under Compliance Review** — churn/deactivation risk, should be cross-referenced against the most-ordered list above; a hero restaurant losing its listing is a materially bigger problem than a long-tail restaurant doing the same

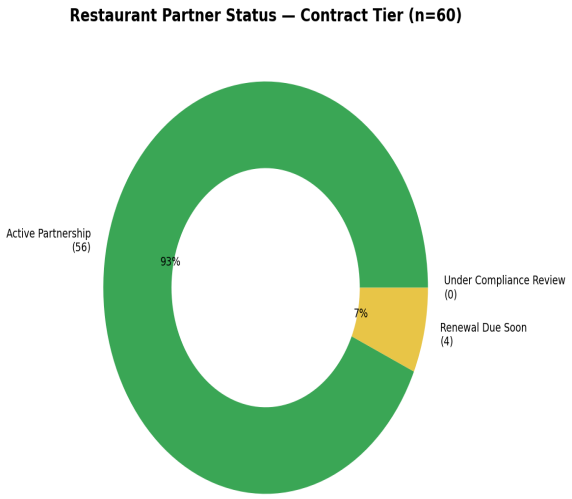


Figure 15. Restaurant partner count by contract tier (n=60).

Key Insight: 4 restaurant partners cross or approach the 30-day renewal trigger — "Momo Palace", "Andhra Bazaar", "Bites Palace" are closest — and none of these overlap with the top-ordered restaurants, meaning current renewal risk is concentrated in lower-profile partners rather than hero restaurants.

7.1 Partners Nearing Contract Renewal — Renewal Risk

The most actionable version of this chart in production cross-tabs contract status against current order velocity, producing a "days until renewal vs. demand" priority score per restaurant rather than a static countdown.

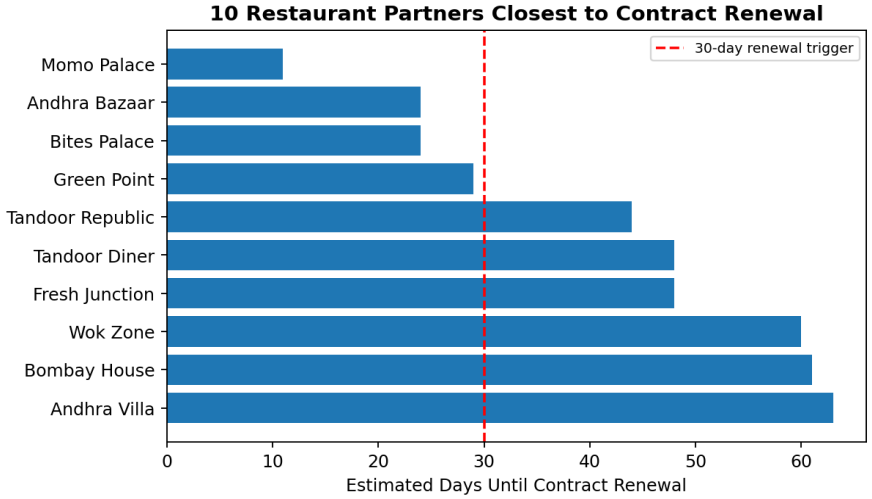


Figure 16. The 10 restaurant partners closest to contract renewal, ranked by estimated days remaining.

Key Insight: Overall takeaway: the biggest cross-cutting risk isn't any single chart — it's the overlap gap: the restaurants at rating risk and renewal risk are currently different from the hero restaurants, but that's exactly the intersection worth monitoring continuously, since a hero restaurant sliding into either category would be materially more damaging than a long-tail restaurant doing the same.

8. Summary of Key Metrics

Metric	Value (simulated)
Total orders (30 days)	~20.3M
Total gross order value (30 days)	~Rs. 577 Crore
Average restaurant rating	3.99 / 5
Top single-day restaurant	94,675 orders (Cafe Bazaar)
Cities tracked	8
Restaurants at Renewal Risk	4 of 60
Peak ordering day	Saturday
Leading device	Android App (52% of orders)
Highest-cancellation cuisine	Biryani (~9.5%)
Deepest Swiggy One adoption	Bangalore (41%)

9. Next Steps to Move From Descriptive → Predictive

This report answers what happened, using the Excel → SQL → Python/Jupyter → matplotlib workflow described in Section 2. The natural next layer builds directly on that same stack, without needing new infrastructure:

- **Demand forecasting per restaurant/city**, using statsmodels or Prophet inside the same Jupyter Notebook, run against the SQL-aggregated daily tables already built for this report, to inform restaurant-onboarding and dark-kitchen decisions.
- **Anomaly detection on rating drops or order spikes**, using rolling z-scores / control limits computed directly in pandas on the existing 7-day rolling-average columns — no new data pipeline required.
- **Cohort and retention analysis**, using pandas groupby on customer-first-order-date cohorts from the same Excel/SQL export, to track repeat-ordering and churn behavior rather than just order volume.
- **Scheduled refresh**, wrapping the notebook in a simple script (cron / Task Scheduler) so the SQL query, pandas aggregation, and matplotlib charts regenerate automatically each time a new Excel export lands.

Happy to build out any of these next, or swap in a real dataset (a live Excel/CSV export or a direct database connection) if one becomes available — the same notebook, SQL queries, and chart structure above would apply directly.

Appendix A. Simulation Methodology

For transparency, the synthetic dataset behind this report was generated as follows:

Parameter	Setting
Restaurants	60 restaurant partners across 10 cuisines
Cities	8 metro demand zones (Bangalore, Delhi NCR, Mumbai, Hyderabad, Pune, Chennai, Kolkata, Ahmedabad)
Window	30 consecutive days
Popularity distribution	Log-normal per restaurant (produces hero-restaurant / long-tail effect)
Seasonality	Day-of-week multiplier — weekend peaks, mid-week dip
Rating drift	Deterministic subset of restaurants drifting downward to simulate a service-quality signal
Ratings	Per-restaurant baseline ~N(4.05, 0.35), with a deterministic subset drifting downward
Partner contracts	Modeled as days-until-contract-renewal, not a static status flag

As with the source methodology this report is modeled on, no real Swiggy order or customer data was accessed or used — all figures above are synthetic and illustrative of the analytics structure only.

Appendix B. Cuisine Reference Table

Cuisine	Orders (30-day)	Revenue (INR)
Pizza	3,351,542	Rs. 1,407,647,640
Chinese	3,665,556	Rs. 953,044,560
Fast Food	3,412,563	Rs. 819,015,120
Healthy Food	1,674,000	Rs. 502,200,000
Biryani	1,380,744	Rs. 469,452,960
North Indian	1,486,436	Rs. 460,795,160
Beverages	2,498,794	Rs. 349,831,160
Rolls & Wraps	1,279,451	Rs. 230,301,180
South Indian	723,436	Rs. 159,155,920
Desserts & Bakery	728,729	Rs. 138,458,510

Key Metrics Snapshot

Metric	Value (simulated)
Total orders (30 days)	~20.3M
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